MIL-B-37246
15 July 1975

MILITARY SPECIFICATION

BRACE, BIT, BONE, CRANIAL, HUDSON, AND BURS, DRILL, AND EXTENSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cranial bone bit brace of the Hudson design and associated burs, drill and extension used in neurological surgery.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Set-up.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

No. 20027	Brace, Bit, Bone, and Component Parts

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Requirements. The brace, burs, drill and extension shall be in accordance with the applicable requirements of GG-I-526 and the following option exercises under that specification:

3.2 Material.

3.2.1 Brace. The bone bit brace (NSN 6515-00-515-2113) shall be fabricated from class 3, types 302, 303 or 304 or class 4, type 420 material.

3.2.2 Extension. The extension (NSN 6515-00-312-4260) shall be fabricated from class 3, types 302, 303 or 304 or class 4, type 420 material.

3.2.3 Burs. The four burs (NSN 6515-00-312-4125, NSN 6515-00-312-4130, NSN 6515-00-515-2114 and NSN 6515-00-515-2115) shall be fabricated from class 4, type 420F material, except that the minimum carbon content shall be 0.30 percent.

3.2.4 Drill. The drill (NSN 6515-00-515-2116) shall be fabricated from class 4, type 420F material except that the minimum carbon content shall be 0.30 percent.

3.3 Hardness. Hardness shall be as indicated for the following types of steel:

Type 302, 303, or 304	Rockwell R_B 75-100
Type 420	Rockwell R_C 40-45, after proper stress relieving.
Type 420F	Rockwell R_C, 50-55, after proper stress relieving.

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3.4 Style, design, and dimensions. Shall be in accordance with DPSC Drawing No. 20027, as indicated for appropriate stock number, and as follows:

NSN 6515-00-312-4125. Shall be a cranial bur of the Hudson design. Bur shall have 9 flutes, 14 millimeters (mm) in diameter. The bur shall be suitable for use with NSN 6515-00-515-2113 (Hudson Cranial Brace) in enlarging trephine openings in the skull.

NSN 6515-00-412-4130. Shall be a cranial bur of the Hudson design. Bur shall have 8 flutes, 20 mm in diameter. The bur, shall be suitable for use with NSN 6515-00-515-2113 (Hudson Cranial Brace) in enlarging trephine openings in the skull.

NSN 6515-00-312-4260. Shall be an extension attachment for use with cranial drill brace of the Hudson design, NSN 6515-00-515-2113. Extension shall be furnished with a set screw for fastening the burs and drill specified herein. The proximal end of the extension shall be designed to fit into the chuck of the drill brace specified above. Extension shall be suitable for surgical procedures involving the human skull.

NSN 6515-00-515-2113. Shall be a cranial drill brace of the Hudson design, suitable for use in surgical procedures of the human skull. Brace shall be constructed to adequately accommodate the burs, drill, and extension specified herein. The brace shall be furnished with a set screw for fastening the burs, drill and extension into the chuck of the brace.

NSN 6515-00-515-2114. Shall be a cranial bur of the Hudson design. Bur shall have 6 flutes, 9 mm in diameter. The bur shall be suitable for use with NSN 6515-00-515-2113 (Hudson Cranial Brace) in enlarging trephine openings in the skull.

NSN 6515-00-515-2115. Shall be a cranial bur of the Hudson design. Bur shall have 8 flutes, 16 mm in diameter. The bur shall be suitable for use with NSN 6515-00-515-2113 (Hudson Cranial Brace) in enlarging trephine openings in the skull.

NSN 6515-00-515-2116. Shall be a flat, cranial perforating drill of the Cushing design. The drill shall be suitable for use with NSN 6515-00-515-2113 (Hudson Cranial Brace) in trephining the skull.

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3.5 Finish. Finish shall be mirror, or mirror and satin.

3.6 Passivation. All material shall be properly passivated.

3.7 Identification marking. In addition to GG-I-526 requirements, items specified herein of foreign manufacture shall also be marked "stainless".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Shall be in accordance with GG-I-526.

4.1.2 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resisting steel alloys may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one instrument.

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Table I. Sampling Criteria for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling Criteria for Tests

Component (Unit of product)	Characteristic	Requirement	Test procedure	Insp level	AQL
1-Bur	Hardness	3.3	GG-I-526	S-2	1.5
or 1-Drill	Dimensions	3.4	------	S-2	2.5
or 1-Extension	Copper sulfate test (For class 4 material)	GG-I-526	GG-I-526	S-2	1.5
or 1-Brace	Boil test (For all material)	GG-I-526	GG-I-526	S-2	1.5

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the manufacture of the burs, drills, extensions, and braces shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of instruments. The items specified herein shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any items in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Item not of the design specified for the indicated stock number.
102	Finish not smooth.
103	Brace not furnished with set screw.
104	Extension not furnished with set screw.
105	Flutes of burs not well defined.
106	Drill point not well defined.
107	Item not free from cracks, holes, pits, burrs, scale and corrosion.
108	Brace does not manually rotate freely and easily.
Minor	
201	Item not identified as specified in 3.7.

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages as shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each brace, burr, drill and extension shall be thoroughly cleaned and wrapped in neutral tissue or other suitable anticorrosive material. The brace shall be packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be utilized. Each bur, drill and extension shall be individually packaged in accordance with MIL-P-116, method IC-1. A preservative shall not be applied to the burs, drill and extension.

5.1.1.2 Intermediate package. An intermediate package shall not be required for the brace or the extension. Six unit packages containing burs of one type only or twelve unit packages containing drills shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty four unit packages containing braces; forty eight unit packages containing extensions; twelve intermediate packages (144 units) containing drills; or twenty-four intermediate packages (144 units) containing burs of one type only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

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5.2.2 Level B. Braces, burs, drills and extensions shall be packaged as specified in 5.2.1 except that the exterior container shall conform to PPP-B-585, class 1. style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brace, burs, drill and extension are to be used in surgical procedures on the human skull.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title and date of this specification.
(b) Applicable National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-312-4125	BUR, CRANIAL, Hudson, 14 mm.
6515-00-312-4130	BUR, CRANIAL, Hudson, 20 mm.

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National Stock Number (con't)	Item Identification
6515-00-312-4260	EXTENSION, BONE BIT BRACE, Cranial, Hudson, 4 inch
6515-00-515-2113	BRACE, BIT, BONE, Cranial, Hudson
6515-00-515-2114	BUR, CRANIAL, Hudson, 9 mm.
6515-00-515-2115	BUR, CRANIAL, Hudson, 16 mm.
6515-00-515-2116	DRILL, FLAT, CRANIAL, Cushing, 3/8 inch.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1960
is

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-37246 BRACE BIT, BONE, CRANIAL, HUDSON, AND BURS,

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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